MARINNA HEICHBERGER

mrh4283@rit.edu (716) 270-7878

EDUCATION

Rochester Institute of Technology, Rochester, NY
Bachelor of Science in Electrical Engineering, expected May 2026
Masters of Science in Electrical Engineering, expected December 2026
GPA: 3.56/4.0

SKILLS

Statistical Learning & Detection: Fisher's LDA, QDA, Gaussian Classifiers, Bayesian Decision Theory, kNN, PCA
Deep Learning: Transformers, Multi-Head Attention, CLIP, CNNs, Graph Neural Networks
Signal Processing & Stochastic Modeling: Fourier Analysis, Spectral Estimation, FIR/IIR Filtering, Random Processes, Sampling Theory, Time–Frequency Analysis
Programming, Embedded & Systems: Python, PyTorch, Hugging Face, PyTorch Geometric, MATLAB, C/C++, NumPy, scikit-learn, Git, Verilog (RTL), FSM Design, FPGA & Embedded Systems

PROJECTS/LABS

Statistical Pattern Recognition & Detection

- Implemented LDA, QDA, kNN, and Gaussian classifiers from first principles.
- Derived and visualized Bayesian decision boundaries under shared and class-specific covariance assumptions.
- Evaluated classifier robustness under class imbalance and covariance shifts.
- Compared classical statistical models to transformer-based architectures for classification tasks.

EEG-Based Cognitive State Classification

- Developed machine learning pipelines for EEG-based drowsiness classification using temporal signal features and sliding-window analysis.
- Evaluated cross-subject generalization challenges under noisy and nonstationary physiological conditions.
- Investigated the impact of preprocessing, normalization, and subject variability on classification robustness.

MSD Data Sonification Windchime

- Designed and integrated a hybrid embedded/software system that converts user-selected and weather-driven musical configurations into physical wind chime actuation.
- Assisted in hardware selection and system-level component specification, including DC motors, encoders, motor drivers, and DC-DC power converters based on electrical and mechanical system constraints.
- Developed the primary Python backend architecture, coordinating GUI interaction, multithreaded execution, JSON state management, weather API integration, and hardware communication workflows.
- Implemented embedded control logic and UART communication protocols between Raspberry Pi and Raspberry Pi Pico systems, including buffered serial handling and edge-case fault management.

Employment Experience

Symbotic LLC – Boston, Massachusetts May 2025- August 2025
Industrial Controls Engineer
Designed modular control architectures for automated material handling systems, emphasizing reliability and system level behavior.
Analyzed system failure modes and implemented logic modifications to improve operational robustness.

Bridgestone Americas - Nashville, Tennessee June 2024 – December 2024
Electrical Engineering Intern

- Developed and deployed real-time fault detection logic for automated tire handling systems, reducing unplanned downtime.
- Programmed sensor-driven control systems using VFDs and photoelectric sensors for autonomous spacing and detection.
- Interpreted electrical schematics to analyze signal flow and ensure correct sensor–actuator integration.

LEADERSHIP/INVOLVEMENT

Rochester Institute of Technology – Rochester, New York August 2023 - Present
Vice-Chair IEEE

- Organized hands-on workshops and networking events to strengthen technical engagement and professional development among electrical engineering students.

Circuits 1 TA

- Lead laboratory instruction and assist students with circuit analysis, instrumentation, and debugging.